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A Multiple Evanescent White Dot Syndrome–like Reaction to Concurrent Retinal Insults

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Purpose: To describe a clinical picture resembling classic multiple evanescent white dot syndrome (MEWDS) potentially triggered by previous or concurrent, apparently unrelated, ocular events and to provide a literature review of similar presentations.

Design: Retrospective chart series and literature review.

Participants: Consecutive patients diagnosed with MEWDS at the Feinberg School of Medicine, Northwestern University, Chicago, Illinois, and the IRCCS San Raffaele Scientific Institute, Milan, Italy, between July 2019 and June 2020.

Methods: Charts of patients were reviewed. Ophthalmic history, best-corrected visual acuity, spectral-domain OCT results, OCT angiography results, fundus autofluorescence results, ultra-widefield fluorescein angiography results, and indocyanine green angiography results were collected. A PubMed-based search was carried out for similar presentations using the terms *MEWDS* and *white spot syndromes*.

Main Outcome Measures: An ocular history positive for previous or concurrent ocular events in patients with MEWDS was sought in our cohort and the existing literature.

Results: Five eyes of 4 patients (2 females; age range, 16–81 years) were included. The first eye had a history of bilateral Best vitelliform dystrophy and unilateral choroidal neovascularization. The second eye had angioid streaks complicated by choroidal neovascularization and underwent prior thermal laser photocoagulation. The third eye had a history of high myopia and a scleral buckle procedure for retinal detachment. The fourth patient had bilateral idiopathic retinochoroiditis. We identified 16 case reports from 5 previous publications that support a MEWDS-like reaction to previous ocular insults.

Conclusions: We suggest a MEWDS-like reaction may be elicited by ocular events in a subset of susceptible patients. We hypothesize that damage to the outer retina may play a role in triggering the local inflammatory response. *Ophthalmology Retina 2021;5:1017-1026 © 2020 by the American Academy of Ophthalmology*

Multiple evanescent white dot syndrome (MEWDS), first described in 1983, is a distinct entity defined as a unilateral or, rarely, bilateral self-limiting inflammatory disease, typically characterized by female gender, young to middle-age predilection, and flu-like prodromal symptoms.[1] Fluorescein angiography (FA) and indocyanine green angiography (ICGA) reveal characteristic dots and spots.[2,3] The dots are correlated with linear and vertical aggregations centered at the ellipsoid zone (EZ) on OCT. The spots are larger outer retinal abnormalities (>200 μm) that are hypofluorescent and sometimes confluent on late ICGA[2,3] and correlate with discontinuities or disruption of the EZ on OCT.[4] Foveal granularity, disc edema, and vitreous cells also may be present.[5]

Although this is the classic presentation, reports have described MEWDS occurring in conjunction with other, apparently unrelated, ocular diseases, either concurrently or over the course of follow-up.[6–12] Similar to classic MEWDS, the lesions are evanescent and have a similar appearance on multimodal imaging. However, the patients do not fit the classic clinical findings, the demographics, or the natural course of the disease seen in MEWDS. Herein, we present 4 patients with MEWDS that occurred concurrently or after retinal insult in the same eye. We speculate that the presence of a coexistent retinal disease may not be incidental, but rather a predisposing factor or a trigger for a local inflammatory reaction with clinical features overlapping those of classic MEWDS. Furthermore, we provide a review of the literature for similar presentations.

Methods

All patients were seen at the Feinberg School of Medicine at Northwestern University in Chicago, Illinois, and the IRCCS San Raffaele Scientific Institute in Milan, Italy, between June 2019 and October 2020. The study was exempted from review by the institutional review board and was conducted in accordance with the tenets of the Declaration of Helsinki. Informed consent was not

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